

Monisha Rengaraj

248-710-5968 | monisha98rengaraj@gmail.com | [LinkedIn](#)

Marysville, OH, United States

OVERVIEW

I am a skilled automotive test engineer with expertise in simulation, debugging, flashing, and automation tools, along with proficiency in design and analysis software. Hands on experience in ensuring quality and reliability of software products by identifying / resolving software defects at bench and vehicle.

SKILLS

Coding platforms: C, C++, JAVA, Python, QNX, Linux, HTML | Network Protocols & Test methods: CAN, CAN-FD, LIN, HIL, SIL Simulations & Debug Tools: Vehicle Spy | Development & Management Methods: JIRA, MS Office | Automation Tool: CATS, GitHub5, SQL server, Tableau, Continuous Improvement, Team Collaboration & Leadership, Fluent Communications skills, Business Analytics, Data Management & Analysis, Troubleshooting.

EDUCATION

Master of Science: Information Technology | Lawrence Technological University | Southfield, Michigan, USA | 2023 | **3.81 GPA**

Courses Completed: Python for data analytics and visualization, Management Information Systems, Project Management, Advanced Computer Networking, Database Management Systems, Data Science for Business, Data Mining, Visual Analytics, System Analysis and Design, Managing Global Workforce, Cybersecurity.

WORK EXPERIENCE

Test Engineer | KPIT | Raymond, OH | JAN 2024 – DEC 2024

- QFIL Flashing on different hardware that goes on different variants.
- Appreciated for finding top-quality software bugs of about 80% and fixing rate of 98% while working on analyzing the stability in the connectivity of HondaLink mobile application.
- Developed about 200 test cases for manual testing and about 100 test scripts in automation.
- Have a good hands-on experience in updating the adaptivist test cycle in JIRA, including modifications of test cases and linking defects with test cases.
- In-Vehicle testing experience and collecting logs makers and CAN both in-vehicle and on bench.
- Delivered a 3-month activity on user experience feedback where appreciation was given with CSAT given by customer of 4.55/5
- Found 6 high-priority crash defects while performing automation testing and stress testing.
- Raised defects on JIRA and modified test cases when required
- Experience on working with Android Auto and Apple CarPlay connections including wired and wireless Bluetooth connection.

Software Verification Engineer | CIE.NET Technologies | Warren, MI | SEP 2022 – JUNE 2023

- Tested using infotainment Bench systems and configured Body Control Module / Central Gateway Module on weekly basis.
- Created test plans, submitted results with weekly test reports and validated onboard diagnostic tools / software based on the requirements.
- Achieved better software quality through debug sessions, troubleshooted system bugs based on user perspective by Ad hoc and manual testing.
- Performed border scenario and stress testing to break the system down and root caused the same.
- Led validation efforts for Radio, DAB, Google assistant Service, Navigation, HVAC, SiriusXM, Wireless Android auto, Apple car play, OnStar, Camera, and other safety features.
- Changed the transmission of the Vehicle and tested the Rear-view Camera using Vehicle Spy based on the guidelines for alert system.
- Orchestrated cross-functional meetings, delivering impactful presentations that communicated key issues, resulting in improved collaboration and accelerated project timelines.
- Analyzed and resolved 50+ software issues through rigorous debugging and root cause analysis, ensuring a top-quality product while reducing bug reports by 10%.
- Executed google voice commands in Google assistant Services to regulate the climate control and check the vehicle system functionality.
- Worked on critical safety features like camera to validate the functionality of alert system.
- Performed regression, sanity, HIL & SIL testing using agile methodology.
- Experienced working on Hypervisor, Android, QNX, Android SDK platforms, Qualcomm image flasher.
- Calibrated through command line shell and troubleshoot the Android based infotainment.
- Proficient in using JIRA, RQM (Rational Quality Management), RTC test management tools, Jenkins, FileZilla and Microsoft Office.

ISRO Indian Space Research Organization | Internship | Bengaluru, India | JAN 2020 – MAR 2020

- Implemented image clustering using Python and Machine Learning Techniques (using AWiFS and LISS IV Data Sets) to cluster multispectral data.
- Clustered and classified data into groups of similar color (water, forest, land, irrigation) segments to differentiate and identify using the K-means algorithm and showed how the image looks with two different data sets.
- Used Jupiter notebook software, Python 3.5 (NumPy, scikit-learn, GDAL) language for coding, and QGIS software to analyze spatial images.

BHEL Bharat Heavy Electricals Limited | Internship | Bengaluru, India | MAY 2018 – JUNE 2018

- Learned and configured the functionality of networks, routers, switches.
- Participated along with the senior network analysts in troubleshooting network issues.
- Actively analyzed and identified problems on connectivity issues, interference in the network and malfunctioning of devices on service calls and helped with possible solutions.

PROJECT EXPERIENCE

- Analyzed the avocado prices and sales volume in multiple U.S. markets.
- Used libraries like pandas to read csv file and convert to data frame, manipulate, and analyze data, NumPy to work on multi-dimensional arrays, and matplotlib to plot graphs.
- Analyzed specific types of avocados, cost and variations over years using group by, computed a range of values to a single value using aggregate, displayed specific rows and columns using option context.
- Analyzed data by creating bar graphs and scatter plot.